ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

PURE MATHEMATICS 4027/1

PAPER 1 2 hours 30 minutes

SPECIMEN PAPER

Additional materials:
Answer booklet
Formulae booklet MF 7
Mathematical tables
Non-programmable Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the answer booklet.

Check that all the pages are in the booklet and ask the invigilator for a replacement if there are duplicate or missing pages.

Answer **all** questions.

All working must be clearly shown.
Omission of essential working will result in loss of marks.
Decimal answers which are not exact should be given correct to three significant figures unless stated otherwise. Decimal answers in degrees should be given correct to one decimal place.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.
Non-programmable electronic calculators or Mathematical tables may be used to evaluate explicit numerical expressions.

This specimen paper consists of 6 printed pages.

Answer all questions

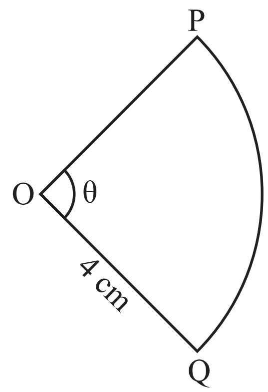
1 **(a)** Simplify $5\sqrt{3}+\sqrt{12}-\sqrt{48}$

leaving answer in surd form. [2]

(b) Solve the equation

$$27^x = \frac{1}{9}.$$ [3]

2



The diagram shows cross section a piece of cake in the form of sector OPQ with centre O and radius 4 cm. The length of arc PQ is 18 cm.

Calculate the

(a) size of angle POQ of the cake, [2]

(b) area of sector OPQ. [3]

3 A stone pillar at Matopo National Park has a shape modelled by the curve $y = x^2$ from $x = 0$ to $x = 10$ metres.

Find the

(a) area under the curve, [3]

(b) volume of the solid formed by rotating the area under the curve about the x-axis. [3]

4 Solve the simultaneous equations,

$2x + y = 3$
$x + xy = 2.$

[5]

5 The pattern shows a Pascal's triangle.

$$\begin{array}{ccccccccc} & & & & 1 & & & & \\ & & & 1 & & 1 & & & \\ & & 1 & & a & & 1 & & \\ & 1 & & b & & 3 & & 1 & \\ 1 & & 4 & & c & & 4 & & 1 \end{array}$$

(a) State the values of a, b and c. [3]

(b) Hence or otherwise expand $(1-5x)^3$ up to and including the term in x^3. [3]

6 The arithmetic progression has first term 4 and third term 8.

(a) Find the common difference. [2]

(b) Given that the first term of a geometrical progression is 5 and the common ratio is $\frac{1}{2}$.

Find the sixth term. [2]

(c) Solve the equation $Cos2\theta = \frac{\sqrt{3}}{2}$, for $0° \leq 0 \leq 180°$. [3]

7 Given that

$$x^3 + 3x^2 - 4x - 12 \equiv (x + a)(x^2 - b),$$

(a) find the value of a and the value of b. [3]

(b) Hence or otherwise solve the equation

$$x^3 + 3x^2 - 4x - 12 = 0.$$ [4]

8 A function f is defined by

$$f : x \rightarrow 6 - \frac{5}{x}.$$

Find

(a) $f^{-1}(x)$, [3]

(b) the values of x for which

$$f(x) = f^{-1}(x)\cdot$$ [4]

9 **(a)** Express $\frac{7x-1}{(x+2)(x-3)}$, in partial fractions. [4]

(b) Write down the first four terms of the sequence defined by

$$U_a = \frac{1}{a^a}$$

[3]

10 $\int_1^4 xe^{-x}dx$

gives the area of Mr Themba's farm.

Using trapezium rule with 4 ordinates, find the approximate area of the farm, giving the answer correct to 3 significant figures. [6]

11 **(a)** Differentiate,

$$f(x) = \frac{2}{3}x^3 - \frac{5}{2}x^2 + 3x.$$ [2]

(b) **(i)** Factorise completely,

$$2x^2 - 5x \quad + 3.$$ [2]

(ii) Hence or otherwise, solve the inequality

$$2x^2 - 5x \quad + 3 < 0.$$ [3]

12 **(a)** Solve the equation

$$log_x 3 = log_3 9$$ [3]

(b) Find the values of x, leaving the answer in exact form.

$$2e^{2x} - 7e^{x} - 4 = 0$$ [5]

13 Samora Machel Avenue (L1) is denoted by the equation

$2y - x = 12.$

Find

(a) **(i)** the equation of Chinhoyi Street which is perpendicular to L1 passing through GPS coordinate $(-1; 3)$. [3]

(ii) the point of intersection of the two roads. [3]

(b) Samora Machel avenue meets the x-axis at A and the y-axis at B.

Find the coordinates of point A and of point B. [2]

14 The position vectors of three bus ranks, Metro (A), Chitima (B) and Croco (C) relative to OK shop (O) are $4\boldsymbol{i} - 9\boldsymbol{j} - \boldsymbol{k}$, $\boldsymbol{i} + 3\boldsymbol{j} + 5\boldsymbol{k}$ and $2\boldsymbol{i} - \boldsymbol{j} = 3\boldsymbol{k}$, respectively.

(a) Find the unit vector parallel to $\overrightarrow{AB}$. [4]

(b) Show that the three ranks A, B and C are collinear. [4]

15 Given

$$f(x) = x^2 - \frac{1}{4},$$

(a) Find

(i) $f(0)$ [1]

(ii) $f(1)$ [1]

(b) Give conclusions from solutions in **(a) (i)** and **(a) (ii)**. [1]

(c) Find $f^{-1}(x)$. [1]

(d) Hence or otherwise use Newton's-Raphson method twice to estimate the root of the equation. $x = x^2 - \frac{1}{4}$.

Start with $x_0 = 0{,}2$. [4]